

On the Effects of Solar Panels on Distribution Transformers

Mohamed A. Awadallah, Tianqi Xu, Bala Venkatesh, *Senior Member, IEEE*, and Birendra N. Singh

Abstract—Energy policies worldwide are mandating large-scale integration of solar panel (SP) generators with inverters on distribution systems. This causes several SPs to be connected to a distribution transformer. The SP and its interfacing inverter alter the performance characteristics of the transformer. In addition, when new sources and loads are connected to a distribution system, from an asset-management perspective, it is imperative to understand and quantify their effect on distribution system components. This paper presents a two-step study on the effects of SP on distribution transformers via simulation and experiments. In step one, the simulation work quantifies the amount of harmonic distortion caused by SP and associated inverters in distribution transformers considering solar farms and rooftop residential installations. The simulation work uses network topology, load, and generation data of a Canadian utility. Various inverter technologies, output powers, carrier signal frequencies, filtering techniques, numbers of active inverters, and transformer configurations are studied. In step two, level of harmonic distortion observed in simulation is created in a laboratory environment using a commercial inverter for SP applications. A three-phase dry-type transformer is tested to observe the effect of higher harmonic distortion on core and winding temperatures of the transformer. Experimental results conclude that under the worst case loading scenario (i.e., full load with active power flow reversed), the transformer lifetime expectancy is anticipated to decrease by 8.3%.

Index Terms—Distribution transformers, harmonics, inverters, lifetime expectancy, photovoltaic, solar panels.

I. INTRODUCTION

RENEWABLE sources, such as solar, wind, tidal, geothermal, biomass, and wave powers provide the rescuing solution to all energy problems of the world as fossil fuels come to an end. Solar energy comes next to wind as the

fastest-developing renewable source [1]. Driven by incentives and ease of installation, rooftop solar panels (SPs) are becoming popular in both downtown and suburban residential areas worldwide. Small rooftop installations of SP produce power in the kilowatt range, while large solar farms can generate several megawatts. In both cases, inverters are used for power conversion from dc to ac. SP sources in this class are usually connected to distribution systems. The use of inverters always causes voltage distortion and injects current harmonics into the system. In addition, connected SPs introduce additional sources on distribution system lines, which reduce power flow from traditional supplies and can cause reverse power flow.

The immediate effects of SP and their associated inverters would be on the distribution transformer, which is designed to supply linear loads at a rated frequency. Voltage and current harmonics are expected to yield extra core and winding losses and, hence, higher temperature rise. If the temperature at the winding hottest-spot increases above a normal value, insulation degradation will be possible, leading to a shortened lifetime and earlier breakdown of the transformer.

The study of harmonics in terms of sources, effects, filtration, and mitigation has been constantly attracting the attention of electrical power system researchers. The effect of power system harmonics on the performance and lifetime of distribution transformers is addressed in many publications [2]–[14]. A calculation routine of the reduced per-unit load at which distribution transformers maintain full lifetime under current harmonics is presented in [2]. Harmonic effects on transformer losses are calculated under current distortion, and a loading profile is proposed in order not to compromise lifetime expectancy [3]. The influence on winding hottest-spot temperature and transformer lifetime, as a result of power loss increase, is evaluated in [4]. Moses *et al.* [5] study the effects of harmonics generated by the plug-in electric vehicles' charging station on the feeding distribution transformer. It is recommended to either derate the transformer operating in such conditions or to install a power-quality (PQ) compensation device, such as active filter, to mitigate harmful effects.

Voltage and current harmonics caused by various nonlinear lighting loads are presented in [6] and the de-rating of the distribution transformer is accordingly proposed. In [7], the finite element method is used to calculate the hottest-spot temperature through field strength solution at different parts of the transformer. The change in the temperature rise due to harmonics is used to estimate the lifetime expectancy, and a new loading profile to keep lifetime unaffected is proposed. Load losses are measured for a five-limb converter transformer at different supply frequencies and converter firing angles [8]. New stray-load and

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eddy-current loss factors are proposed in terms of harmonic frequency and firing angle. A laboratory setup is proposed to measure losses of high switching frequency converter transformers, and compare to those computed through finite elements [9].

An inverter is used to feed the transformer, which is loaded with nonlinear loads in order to generate current harmonics [10]. The change in losses under such condition is evaluated through short-circuit and open-circuit tests. In [11], electrical and thermal parameters of distribution transformers, including top-oil and hottest-spot temperatures, are assessed in a harmonic environment. Measuring core temperatures with resistive and nonlinear loads, it is found that current total harmonic distortion (THD) of 20% and 60% result in increase of core temperature of 20% and 50%, respectively [12]. Choudhury *et al.* [13] examine the effect of supply voltage distortion on transformer excitation current, where the phase angles of voltage harmonics have more impact than the magnitudes on the corresponding current harmonics. An expansion of the standard K-factor is able to estimate the composite harmonic current caused by multiple linear and nonlinear loads [14].

Effects of solar photovoltaic (PV) panels on the distribution networks are also studied [15]–[18]. The replacement of 30% of the incandescent lamps by compact fluorescent lamps, for the sake of energy saving, results in 8% voltage THD in a weak network supplied by SP [15]. Voltage THD may reach 31% if the replacement is for 90% of the lighting load. Voltage and current THD of the accommodating network remain within standard limits even if all SP of the Sydney Olympic Village are simultaneously functioning [16]. Impact of SP on voltage harmonics of power networks at two Greek islands is addressed in [17], and compared to the case of Diesel generator supplies. In [18], the effect of solar panels and associated inverters on the power quality of distribution networks and transformers is investigated. Cases of unbalanced operation of distribution transformers are studied through simulation and experimentation.

Upon integrating a PV system to a power grid, the harmonic distortion of voltage and current is mainly caused by the inverter. Rigorous research efforts have been recently devoted to the enhancement of the topology, performance, and control of PV system inverters [19]–[24]. Inherent system redundancy is increased through distributed control of fault-tolerant modular multilevel inverters for PV applications [19]. Output harmonic content of multilevel inverters connected to SPs is minimized through the generation of switching angles by artificial neural networks (ANNs) [20]. In [21], the number of power-electronic components in an SP multilevel inverter is reduced by using a ladder architecture. An integrated dc/dc converter is employed in [22] to interface SP to microgrids via an energy delivery and management system. Maximum power is harvested from SPs at the unity power factor through a battery-assisted quasi-Z-source inverter which controls the SP voltage by introducing shoot-through states [23]. For a single-phase grid-tied PV system, constant power operation and maximum power tracking are concurrently achieved via a boost converter and line-commutated inverter as an interface [24]. Fixed firing angles of the inverter keep the dc-link voltage constant, while maximum power is absorbed through duty ratio control of the boost converter.

It appears that literature work separately considers harmonic effects on distribution transformers, SP impact on voltage and current distortion, and the development of inverter technology. Accordingly, research considering the direct impact of SP on distribution transformers, especially in a distributed generation and harmonic-contaminated environment, is missing. Moreover, from an asset-management perspective of utilities, it is imperative to understand the effects of new sources and loads on distribution system components.

In this paper, the effects of voltage and current harmonics caused by SP inverters on the performance of distribution transformers, especially during second quadrant operation resulting from a reversal of active power due to generation exceeding load, are studied in a two-step process. In step one, many technical issues are addressed, including different inverter topologies, carrier frequency, output power, filtering techniques, number of active inverters, and transformer winding configuration. The impact of such factors on harmonic distortion of voltage and current waveforms is studied via simulation of a real distribution system. Step one quantifies the amount of harmonic distortion in distribution transformers under different factors.

In step two, transformer performance is measured in the laboratory under normal and SP operation scenarios. The measurements are used to verify simulation results concerning the impact of output power and winding configuration on harmonic levels. Two reasons are behind the selection of these two factors, in particular, for experimental verification: practicality and interest of utilities. Thereafter, in step two, measured core and winding temperatures are employed to evaluate the lifetime expectancy. Under worst case operating conditions, the transformer lifetime is expected to reduce by 8.3%.

This paper answers two vital questions, which have been asked by the utilities for quite some time. The questions concern the impact on performance characteristics and expected lifetime of distribution transformers due to reverse active power flow and harmonic contamination caused by SPs. Findings of this research warn utilities against a condition where the capacity of the SP connected to a distribution transformer is equal to or exceeds its rating. In case such a condition is inevitable, care should be taken about transformer operation with full load in the second quadrant. Energy-storage elements may be used on the SP side of the transformer in order to avoid full-load reverse power flowing into the distribution network.

II. STEP ONE—SIMULATION AND RESULTS

An assessment of the impact of SP and associated inverters on waveform distortion is first studied. A Matlab/Simulink model for a solar farm and accommodating distribution system (Fig. 1) is built based on actual data of the Canadian utility. The transmission system is modeled as a 170-MVA, 115-kV three-phase supply, and a 5-MVA, 27.6/0.6-kV three-phase transformer is modeled to connect the 100-kW solar farm into the distribution network. The system incorporates a dc converter, employing a maximum power point tracking (MPPT) algorithm, three-phase inverter, and an LC filter as shown in Fig. 1. Voltage and current THD at Buses 1 and 2 are monitored. The effects of SP on

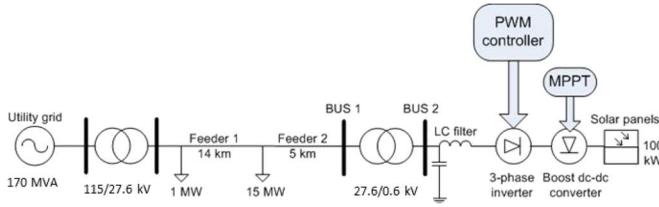


Fig. 1. General model for the solar farm.

TABLE I
VOLTAGE AND CURRENT THD FOR DIFFERENT INVERTER TECHNOLOGIES

Control	Phase-to-ground				Phase-to-phase
Topology	2-level IGBT	3-level IGBT	3-level MOSFET	Double 3-level IGBT	3-level IGBT
V_1	0.123	0.073	0.073	0.043	0.080
I_1	4.717	3.410	3.410	3.263	3.367
V_2	5.953	3.380	3.380	2.923	3.753
I_2	4.703	3.407	3.407	3.257	3.367

waveform distortion are studied concerning different technical issues.

A. Effect of Inverter Technologies

In order to compare the effects of different inverter technologies on the network, three inverter topologies, two control schemes, and two power-electronic devices are used. Two-level, three-level, and double three-level inverters using insulated-gate bipolar transistors (IGBT) and metal-oxide-semiconductor field-effect transistors (MOSFET), and employing phase-to-ground and phase-to-phase voltage control are simulated. The objective is to examine harmonic distortion of voltage and current waveforms under different inverter technologies.

Voltage and current THD results are shown in Table I. It is concluded from total harmonic distortion (THD) values that using an IGBT or MOSFET has the same effect on waveform distortion. It is also noted that phase-to-ground voltage control gives less voltage THD and almost equal current THD to phase-to-phase voltage control. In terms of the inverter topology, the double three level gives less THD than three-level inverters. However, a double three-level inverter employs twice as many devices as a three-level inverter. On the other hand, the two-level inverter yields higher voltage and current THD, which can exceed the 5% IEEE standard limit given in [25]. Accordingly, the three-level topology dominates the industry of three-phase inverters for solar applications.

B. Effect of Output Power

The effects of SP output power, which is dependent on solar irradiation and cell temperature, on waveform distortion are investigated. Fig. 2 shows the variation of THD against SP power. It is apparent that voltage THD on both sides does not change remarkably with output power, and is always within the standard limit of 5% [25]. However, current THD is high at small values of output power, and decreases with increasing power until they go below the standard limit around rated power. It

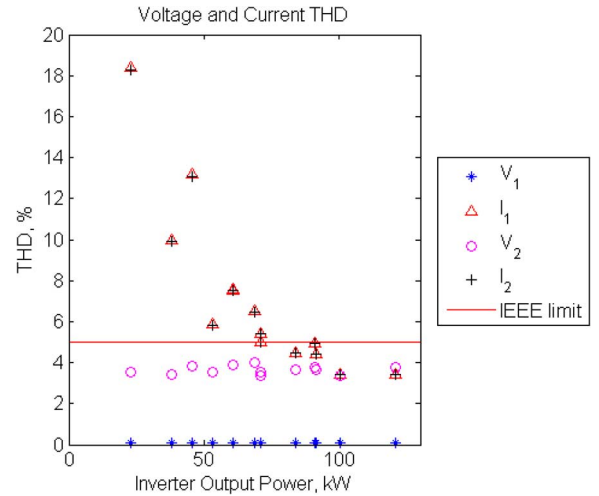


Fig. 2. Variation of THD with output power.

TABLE II
VOLTAGE AND CURRENT AVERAGE THD FOR DIFFERENT SP RATINGS

Rating of SP relative to system capacity	Average voltage THD	Average current THD
1/10000	0	1.06
1/1000	0.02	1.07
1/100	0.17	1.16
1/10	0.74	0.98
1/5	0.89	1.13
1/4	0.96	1.31
1/3	4.33	7.79

is anticipated that voltage THD may not be greatly affected by SP power where the ratio between rated SP output and system power carrying capacity is low, that is, where the system power carrying capacity is several times the rated SP output at the point of common coupling (PCC). Thus, the model is run for different SP ratings, where average voltage and current THD are given in Table II. It is evident that the relative rating of SP, referred to as system capacity, influences the average voltage and current THD. Nevertheless, voltage and current THD seem to have acceptable values for low relative SP ratings as well.

C. Effect of Carrier Signal Frequency

The dominant switching technique in three-phase inverters manufacturing is pulsewidth modulation (PWM). The instantaneous magnitude of a triangular carrier signal is compared to that of a sinusoidal control signal in order to determine the switching logic. The frequency of the carrier signal sets forth the harmonic content of the output waveform of the inverter. It also determines the switching frequency and switching loss of inverter devices, and influences noise and interference with nearby circuits and systems.

The frequency of the PWM carrier signal is varied, while voltage and current THD are monitored and plotted in Fig. 3. It is clear that voltage and current THD on both sides of the transformer decrease as carrier frequency increases. Certain carrier frequencies result in all THD values being below the standard

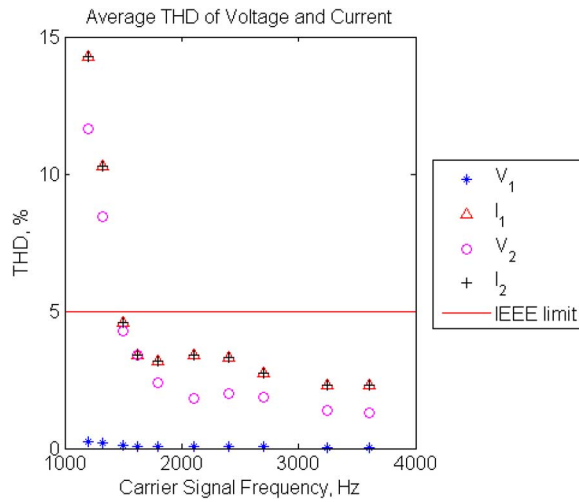


Fig. 3. Variation of THD with carrier signal frequency.

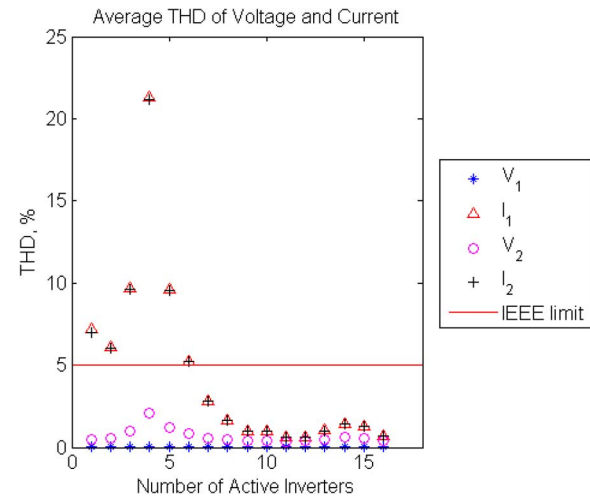


Fig. 5. Variation of THD with the number of active inverters.

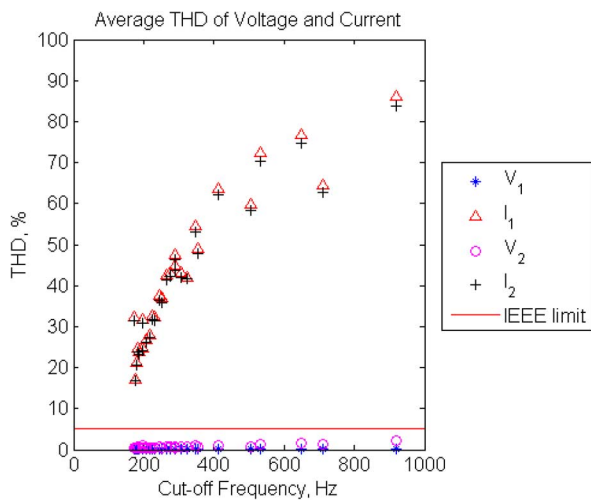


Fig. 4. Variation of THD with cutoff frequency.

limit. However, the carrier frequency should be carefully selected since it affects switching losses. Industrial practices state that IGBTs are dominant in medium power and voltage applications. The switching frequency is then in the low kilohertz range, typically between 3 kHz and 10 kHz.

D. Effect of Filtering Techniques

All inverter manufacturers are keen on designing effective passive filters such that the output of their inverter is minimally distorted. The cutoff frequency of the filter is a vital factor in determining which harmonic components will be filtered out, and deciding on the size and cost of filter elements. A second-order low-pass filter is used in the simulation model as shown in Fig. 1. The cutoff frequency of the filter is varied, while the voltage and current THD are recorded and plotted in Fig. 4.

Voltage THD on both sides of the transformer is always less than standard limit; however, voltage THD decreases as the filter cutoff frequency decreases. Current THD on both sides seems to be more affected by cutoff frequency; they reach a minimum of about 17% as the cutoff frequency becomes

almost three times the power frequency. It should be noted that as cutoff frequency increases, more harmonic components can penetrate through the network. On the other hand, the size, weight, and cost of filter components dramatically increase as cutoff frequency is reduced. In addition, the fundamental component of the output voltage becomes more attenuated as cutoff frequency decreases. Therefore, cutoff frequency of the passive filter should be carefully designed in order to best compromise such merits and demerits. Cutoff frequency is typically chosen at about one-tenth of the device switching frequency. Accordingly, in case of IGBT inverters, the cutoff frequency of the output filter is less than 1 kHz.

E. Effect of the Number of Active Inverters

The single SP and one inverter of the model of Fig. 1 are broken down to 16 pairs of SPs and inverters at a lower rating in order to represent an actual case. The effect of the number of active inverters at a time on the waveform distortion is explored. Voltage and current THD on both sides of the transformer are plotted against the number of active inverters in Fig. 5. Voltage THD is always less than the standard limit, while current THD is higher than the limit when few inverters are active. Current THD becomes a less than standard limit when the number of active inverters is seven or more. However, it should be noticed that a clear sharp peak of current THD on both sides and voltage THD on the secondary side takes place when the number of active inverters is four. The reason is believed to be the resonance between LC filters of active inverters. To validate such reasoning, the filter parameters are changed for all inverters, and the simulation is repeated. Consequently, the peak THD point moves to correspond to six inverters. When filter parameters vary again, the peak THD point occurs at two active inverters. Therefore, filter designers and system operators should avoid this condition because of its harmful reflection on THD values.

F. Effect of Transformer Winding Configuration

The effects of different winding connections and vector groups on waveform distortion are examined, while core saturation is modeled. Table III shows THD values of primary

TABLE III
VOLTAGE AND CURRENT AVERAGE THD FOR DIFFERENT TRANSFORMER CONNECTIONS AND VECTOR GROUPS

	Yg/d-11	Y/d-11	D/yg-11	D/y-11	Yg/d-1	Y/d-1	D/yg-1	D/y-1	Yg/y-12	Y/yg-12	Y/y-12	D/d-12	D/d-10	D/d-2
V_1	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
I_1	8.58	8.58	8.58	8.58	7.19	7.19	7.19	7.19	5.61	5.61	5.61	5.61	4.38	4.45
V_2	0.47	0.47	0.47	0.47	0.45	0.45	0.45	0.45	0.42	0.42	0.42	0.42	0.39	0.39
I_2	8.36	8.36	8.36	8.36	7.01	7.01	7.01	7.01	5.46	5.46	5.46	5.46	4.27	4.33

and secondary voltage and current at different winding configurations. It is evident that the primary voltage THD is almost independent of the transformer winding configuration since it is predominantly controlled by the grid since the grid is strong relative to the SP (ratio of short-circuit levels). Also, it seems that grounding of the neutral point does not affect the voltage or current THD. However, the vector group of winding connections appears to have a significant influence on the harmonic level. If the primary winding of a distribution transformer is delta and the secondary star grounded, the results denote that the vector group yields one less THD than 11.

III. STEP TWO—EXPERIMENTS AND RESULTS

The experimental work aims at testing SP effects on three-phase transformers through a laboratory setup. The effects of reverse active power flow and harmonics generated by SP inverters are to be particularly addressed. The impact of load power change and transformer winding configuration on voltage and current THD levels is shown in order to verify simulation results. Through the experimental study, variations of steady-state core and winding temperatures are tracked in order to ultimately assess the transformer lifetime.

A three-phase grid-tied commercially available inverter for SP applications is used to feed the secondary winding of a three-phase dry-type transformer, whose primary is connected to the grid. The inverter applies THE maximum power point tracking (MPPT) algorithm over two channels, which could be separately fed from two SPS, or paralleled to the same one. The inverter has a second-order passive filter, which helps reduce distortion of the output waveforms. This inverter is selected from different market alternatives since the THD of its output voltage is below the 5% IEEE standard limit [25]. The winding configuration of the transformer varies between D/y-1, D/yg-1, Y/d-11, and Yg/d-11. The dc source supplying the inverter can run in a current mode to best represent the constant current characteristic of SP. The setup is shown in Fig. 6. The secondary-side three-phase inductor maintains reactive power flow from the grid, while active power flows into the grid, in order to signify second-quadrant operation of the transformer.

Two three-phase PQ analyzers are used to measure voltage and current THD on both sides of the transformer as well as active, reactive, and total powers. Core and winding temperatures are also measured via two independent thermocouples, where readings are obtained by a data logging module connected to a personal computer. The transformer is tested under SP operation through the setup of Fig. 6; it is also tested under normal

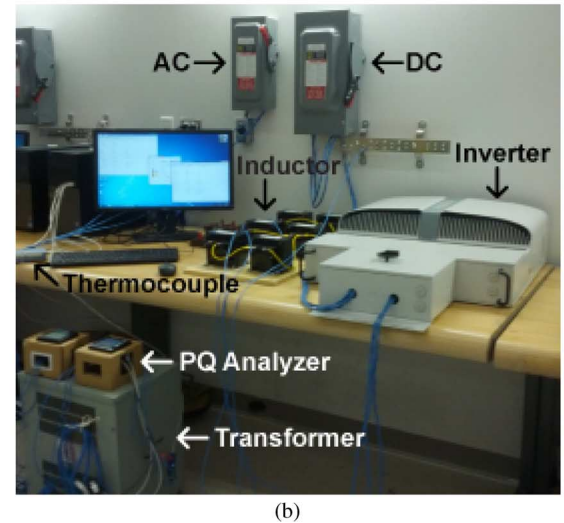
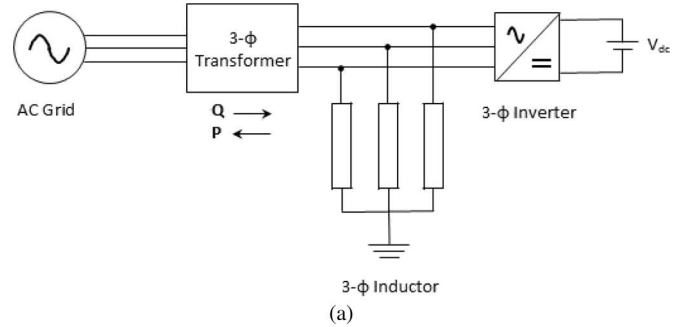


Fig. 6. Experimental setup. (a) Schematic. (b) Picture.

conditions when supplied by a sinusoidal source feeding constant linear load.

The effect of SP power and transformer winding configuration on waveform distortion is tested. Voltage and current THD are measured under normal and SP operations with all possible winding configurations. Voltage and current THD are plotted in Fig. 7 against inverter power for a D/y connection. Voltage THD is always less than the standard limit although it increases with loading under SP operation. This increase in voltage THD with loading is believed to be due to the increasing harmonic drop across the feeder cable and transformer leakage impedance when the distorted load current increases. Current THD under normal operation is always less than the limit. However, under SP operation, current THD reaches a high value at light load, then decreases with loading until it becomes within the standard limit close to full load.

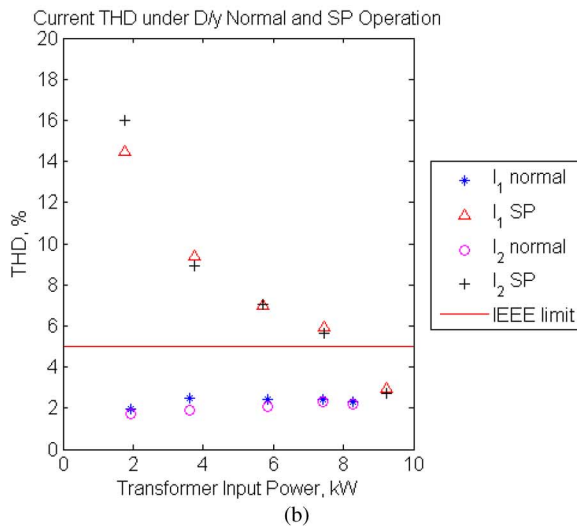
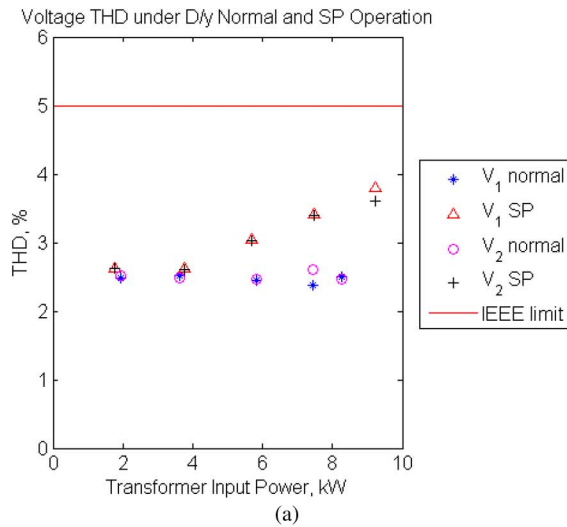


Fig. 7. Variation of THD with loading under normal and SP operation. (a) Voltage. (b) Current.

The primary (grid side) and secondary (inverter side) voltage and current THD under SP operation are shown in Figs. 8 and 9 for Y/d and Yg/d connections, respectively. Voltage and current THD maintain the same trend of change versus SP power. Slight and inconclusive differences exist between corresponding values indicating that grounding of the neutral point does not have a significant influence on waveform distortion. Experimental measurements of voltage and current THD, shown in Figs. 7–9, acceptably match and verify simulation results given in Fig. 2 and Table III.

IV. STEP TWO—COMPUTATION OF LIFETIME EXPECTANCY

The temperature rise of any electrical device or equipment is a key factor determining its lifetime. Distribution transformers are no exception. The winding insulation, which is the weakest thermal point, is normally designed with a specific thermal class such that the temperature at the hottest spot of the winding should not exceed a certain value. The location of the hottest winding spot in a transformer during its operation is normally unknown, and is very difficult to determine (determined only through the postfailure postmortem).

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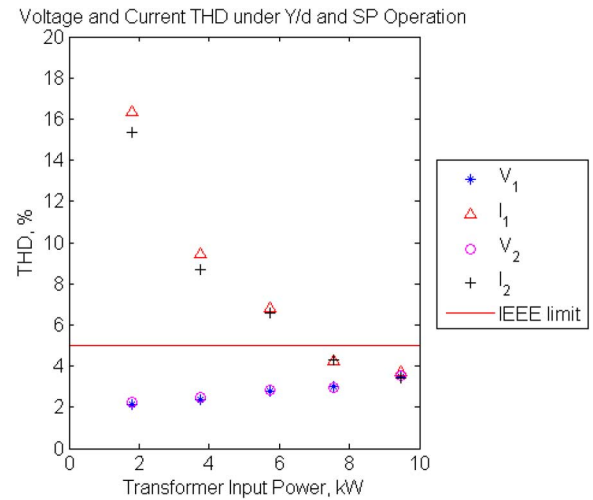


Fig. 8. Variation of THD with loading with the Y/d connection and SP operation.

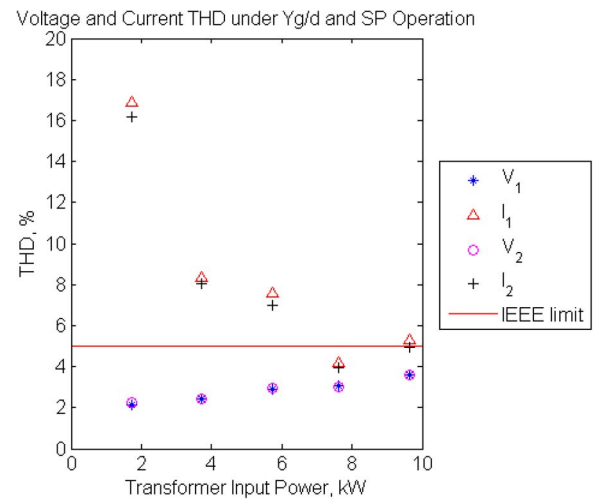


Fig. 9. Variation of THD with loading with the Yg/d connection and SP operation.

The basic loading conditions of a transformer for normal life expectancy are [26] as follows.

- 1) The transformer is continuously loaded at rated kilovolt amperes and rated voltage.
- 2) The average temperature of cooling air during any 24-h period is equal to 30 °C, and the maximum temperature of cooling air at no time exceeds 40 °C.
- 3) The altitude does not exceed 1000 m (3300 ft).

In dry-type transformers, the hottest-spot winding temperature is the direct sum of ambient temperature, average winding temperature rise, and hottest-spot allowance [26]. It is therefore indicated that a change in any of these three components results in an equal change in the temperature of the hottest winding spot. This fact is noteworthy because it sets forth the basis for lifetime computation under SP operation. It is also stated in [26] that for a dry-type transformer with a temperature class of 220 °C, the hottest-spot temperature of the winding has been assumed to be 210 °C if the three basic loading conditions mentioned earlier are maintained (accepted norm and standard). Such hottest-spot temperature is expected

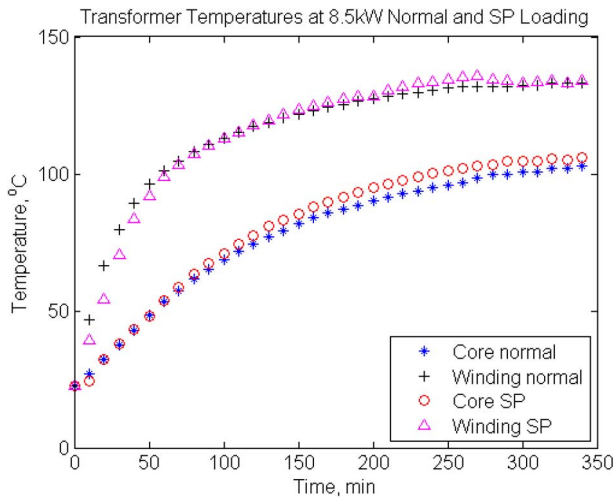


Fig. 10. Variation of transformer temperatures with time at 8.5 kW.

to yield a normal lifetime expectancy of 20 years [26]. It should also be highlighted that the guidelines given in [26] leave 10 °C allowance, or a safety margin, between the design thermal class specification and the operating temperature at the hottest winding spot for any dry-type transformer.

Core and winding temperatures of the transformer under test are measured in separate experiments during normal and SP operation. Temperatures are recorded every ten minutes and are long enough to reach a thermal steady state or close. At 8.5-kW load, Fig. 10 shows the core and winding temperatures under normal (i.e., sinusoidal source without integration of SP) and under SP operation. Winding temperatures seem comparable, while core temperature under SP operation is slightly higher than normal conditions. The reason is the strong dependence of core loss on the frequency of applied voltage. The small increase in core temperature under SP operation indicates slightly higher core loss and lower efficiency.

The same plots are shown in Fig. 11 at the 10-kVA transformer full load with second-quadrant operation. Core temperature under SP operation maintains the same slight increase above the normal operation case. Although winding temperatures appear to be almost equal, careful inspection of the temperature values indicates a steady-state difference of 1.2 °C in favor of SP operation. It is of interest to know the potential impact of such a small increase in winding temperature on the lifetime expectancy of distribution transformers. It should be also noticed that although the load of Fig. 10 is 85% of the load of Fig. 11, the steady-state winding temperature of Fig. 10 is less than 70% of the steady-state winding temperature of Fig. 11. Therefore, the correlation between steady-state winding temperature rise and load level is not linear. Average values of the voltage and current THD during the loading experiments of Figs. 10 and 11 are given in Table IV. It is apparent that waveforms are more distorted at 8.5-kW than the 10-kVA load. However, the effect of harmonic distortion on temperature rise is more significant at the full load of 10 kVA. The reason is that the winding is at its permissible temperature limit under full-load operation so that any consistent increase in temperature would directly affect the transformer lifetime.

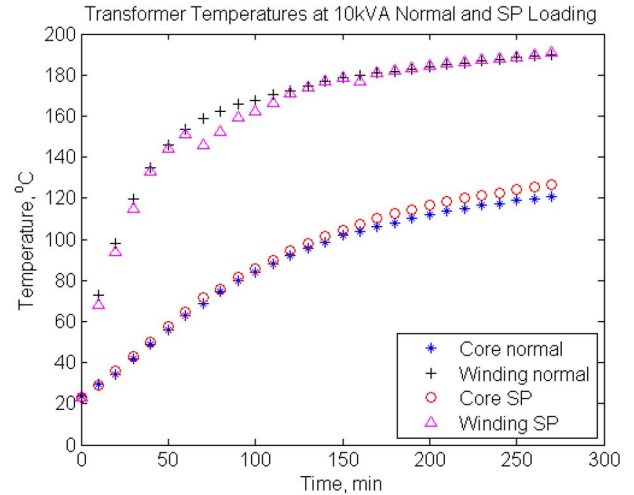


Fig. 11. Variation of transformer temperature with time at 10 kVA.

TABLE IV
AVERAGE THD AT 8.5-kW AND 10-kVA LOADING CONDITIONS

Condition	8.5 kW		10 kVA	
	Normal	SP	Normal	SP
I_1	2.15	5.16	1.64	3.58
I_2	2.17	4.90	1.52	3.72
V_1	2.15	3.43	2.09	3.32
V_2	2.16	3.42	2.11	3.25

The winding temperature curves of Fig. 11 are of particular significance since they constitute the basis for lifetime computation under SP operation. It should be emphasized that the winding temperature measured in the laboratory and plotted in Fig. 11 cannot be claimed to be the hottest-spot temperature. Nevertheless, the change in such measured temperature, from normal to SP operation, indicates an equal change in the hottest spot of the winding as indicated in [26]. Since the transformer under consideration is of the 220° class, it is accordingly assumed that the hottest-spot temperature at nominal load and voltage is 210 °C [26]. The reason behind such an assumption is that the basic loading conditions for normal life expectancy mentioned earlier are maintained during the normal operation case of Fig. 11. The hottest-spot temperature under SP operation is therefore 1.2 °C higher than the normal case, according to the measurements. For a 220 °C temperature class dry-type transformer, the lifetime expectancy in hours is given in [26] in terms of the absolute hottest-spot temperature T as

$$\text{Life} = 10^{\left(\frac{2582}{T} - 10.453\right)}. \quad (1)$$

According to the results and testing procedures of this PAPER as well as the guidelines given in [26], a systematic mechanism to compute transformer lifetime under SP operation could be formulated as follows.

- 1) Load the transformer from a sinusoidal power supply, maintaining the basic loading conditions for normal life expectancy.
- 2) Measure the winding temperature at any point until thermal steady state is reached.

TABLE V
EQUIPMENT SPECIFICATIONS OF THE EXPERIMENTAL SETUP

Transformer	
Rating	10 kVA
Voltage	240/240 V
Connection	Δ/Y
Cooling	ANN
Temperature Class	220 °C
Frequency	60 Hz
Impedance % at 170 °C	4
Enclosure	NEMA-3R
Inverter	
Rated Grid Voltage	208 V
Rated Power	10 kW
MPPT Channels	2
Maximum Power per Channel	6800 W
Maximum Current per Channel	24 A
Maximum Efficiency	96.5%
Start-up Voltage	120 – 350 V
Cooling	Natural Convection
Enclosure	NEMA-4X
Inductor	
Rated Current	20 A
Inductance	10 mH
Thermocouple	
Number of Channels	8 differential
A/D Converters	4 dual 24-bit, Sigma-Delta type
Differential Input Voltage Range	± 0.08 V
Input Impedance	5 G Ω min
Resolution	24 bits
Warm-up Time	30 minutes min
Power Quality Analyzer	
Voltage Resolution	0.1 V
Voltage Range	20 V – 1000 kV
Current Resolution	0.1 mA
Current Range	5 mA – 50 kA
Frequency Resolution	0.01 Hz
Frequency Range	45 – 65 Hz
Samples per Cycle	32
Harmonics Range	1 – 16

- 3) Assume that the hottest spot winding temperature is equal to the transformer thermal class minus the 10 °C allowance.
- 4) Load the transformer, in a separate experiment, from an SP supply maintaining the basic loading conditions for normal life expectancy.
- 5) Measure the winding temperature at the same point until thermal steady state is reached.
- 6) Find the steady-state temperature difference between SP and normal operations.
- 7) Add the temperature difference to the normal hottest spot temperature, of step 3, in order to estimate the winding hottest spot temperature under SP operation.
- 8) Substitute the hottest spot temperature of SP operation into (1) to compute lifetime expectancy.

Therefore, substituting T as 483 °K, standing for 210 °C, in (1) yields a lifetime of 20 years, which is considered the expected lifetime under normal full-load operation as given in [26]. However, when T is 484.2 °K, standing for the normal 210 °C plus the 1.2 °C rise due to SP operation, lifetime expectancy becomes 18.336 years. Thus, lifetime expectancy of the transformer is anticipated to reduce by 8.3% if continuously operated in the second quadrant at full load with the SP inverter. This is a key finding of this paper and has long nagged electrical distribution utilities that experience power-flow reversals and harmonic

distortion on their systems due to high penetration of renewable generation, such as solar and wind.

V. ASSUMPTIONS AND LIMITATIONS

The results presented in this paper are based on some assumptions in simulation and experimentation. The simulation model assumes balanced three-phase operation; cases of imbalance are not considered. Modeling of the inverter assumes identical power-electronic switches and neglects switching losses. Passive components of the filters are assumed ideal by disregarding inductor and capacitor losses. Sinusoidal PWM is always used for inverter switching. Transformer loading tests are conducted for a period of time long enough to reach steady-state temperature. On different transformer testing experiments, temperatures are measured at the exact same points of the core and winding.

The results of lifetime reduction are generally limited to dry-type transformers; oil-immersed transformers have different loading guidelines and lifetime calculations [27]. The expected lifetime reduction of 8.3% is true for dry-type transformers of the 220 °C thermal class, which experience waveform distortion as quantified in this paper, leading to a consistent increase of the hottest winding spot by 1.2 °C above normal on the steady state. For other transformer design specifications or loading conditions, steady-state temperature has to be compared between normal and new operations. The procedural steps mentioned earlier have to be followed to evaluate the anticipated change in transformer lifetime. Since the inverter is the main source of harmonics and the major cause of temperature rise above normal operation, care should be taken about different inverter designs, modulation techniques, and control schemes.

VI. CONCLUSIONS

This paper addresses key concerns from electrical utilities with respect to operation of power transformers under conditions of reverse flow and harmonic distortion triggered by renewable generation, such as solar and wind. A two-step study of the electrical and thermal effects of solar panels on distribution transformers is presented. In step one, a Matlab/Simulink model is built for a solar farm and its accommodating distribution system based on Canadian utility data, including network topology and parameters. Different factors affecting waveform distortion across the distribution transformer are considered.

The three-level inverter topology employing phase-to-ground voltage control is an excellent selection compromising moderate waveform distortion with design complexity. However, the type of power-electronic device to build the inverter does not have much influence on voltage and current THD. Voltage THD maintains acceptable values below standard limits, whereas current THD is high at light loads and decreases with loading. Waveform distortion increases with the SP rating in reference to system capacity. Carrier frequency of sinusoidal PWM and cutoff frequency of the passive filter needs to be carefully designed due to their impact on THD values and other performance characteristics.

In case the solar farm rating is so high that many inverters are deployed, care should be taken on the number of active inverters at a time. Resonance between a certain number of filters could give rise to abruptly high THD values. The transformer winding configuration has some impact on waveform distortion, especially the vector group of the winding. Step one of this work gives an insight on the harmonic distortion of voltage and current waveforms across a distribution transformer due to SP and associated inverters.

In step two, transformer performance is measured in the laboratory under normal and SP operating conditions. Experimental results, recorded during step two, match simulation findings of step one with respect to the effects of SP output power and winding configuration on waveform distortion. Under the worst case loading scenario of full-load second-quadrant SP operation, the winding hottest-spot temperature would be 1.2 °C higher than normal conditions. Such a small increase in temperature rise can yield a reduction of 8.3% in the lifetime expectancy of distribution transformers. This finding is critical to utilities experiencing increasing penetration of SPs into distribution networks. In order to maintain normal lifetime expectancy, distribution transformers must not be connected to SP of equal or higher aggregate capacity. Otherwise, energy-storage devices may be installed in order to prevent harmonic-distorted second-quadrant operation of the transformer at full load. It should be highlighted that energy policies in many jurisdictions worldwide allow up to 10-kW PV rooftop installation per residential customer. This has a strong potential to overload distribution transformers with reverse flows under harmonic distortion.

The contribution of this paper is to quantify the different sources of harmonics affecting the performance of distribution networks and transformers as a result of integrating SP generators. Then, such quantification is employed in a laboratory setup to assess the reduction in lifetime expectancy of a distribution transformer subject to reverse power flow and harmonic distortion caused by SP integration. A well-defined set of procedures to determine changes in transformer lifetime expectancy due to variations in loading scenarios is also given.

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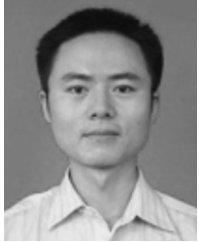
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